

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Piping Routing Study and Stress-Critical Line List

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CFU-000-PI-RPT-001 Piping Routing Study and Stress-Critical Line List

1 Purpose and scope

FEED piping routing study for the 100 kBPSD crude + vacuum distillation unit. The study establishes the pipe-rack and equipment piping arrangement in a 3D orthogonal routing model (data/routing.json, 3D model CFU-000-PI-3DM-002, viewer viewer-piping.html), issues isometrics for the layout- and stress-critical lines, quantifies rack loading, lists stress-critical lines with simplified flexibility checks, and derives pipe wall thicknesses and the MTO (CFU-000-PI-MTO-001). All inputs are read at build time from data/lines.json (line list), data/layout.json (plot plan, nozzles, rack), data/instruments.json, data/psv.json and data/mech.json.

- Lines in line list: **245**; routed: **182** (13 critical/isometric, 11 unit headers, 158 study-level auto-routed); not routed: **50** (reasons in section 4 and MTO sheet Line-List-Crosscheck).
- Routed pipe 11,813 m net (12,956 m incl. allowance), 1,486 fittings, 428 flanges, 4,448 butt welds (25,828 shop + 11,720 field inch-dia), 1,359 spools, 2,153 supports.

2 Codes and references

- ASME B31.3-2022 Process Piping (wall thickness 304.1.2, flexibility 319, testing 345, PWHT 331, NDE 341)
- ASME B36.10M, B16.5, B16.47 Series A, B16.9, B16.10, B16.20; MSS SP-58/69 supports
- API 610 12th ed. (pump nozzle loads, Table 5), API 560 (fired heater terminals), API 661 (air-cooler nozzles), API 521 (relief/flare piping), WRC 537/297 (vessel nozzle local stresses)
- CFU-000-PI-SPC-001 Piping class summary; CFU-000-PL-PLT-001 plot plan; CFU-000-PL-ELV-003 rack section; CONVENTIONS.md (rack tiers EL 106.0 / 108.5 / 111.0, bents at 6 m)

3 Routing philosophy

- **Orthogonal routing** nozzle -> stub -> drop/rise -> rack tier -> along rack -> off rack -> destination nozzle. Lines change elevation to change direction ('elevation by direction'): local north-south pipeways at EL 104.0, east-west at EL 105.4; exchanger-area pipeway EL 104.6.
- **Rack tier allocation:** tier 1 (EL 106.0) hot process B2/B3, crude, residue, pumparounds (T >= 260 C); tier 2 (EL 108.5) products, cold hydrocarbon, sour water; tier 3 (EL 111.0) utilities, steam, condensate, CW, fuel gas, flare. Hot / loop-bearing lines are placed at the rack edges so loops span the rack width; heaviest lines next to the bent columns.
- **Rack entry/exit:** tier-1 lines leave the rack below the tier (clear of tier-1 loops), tier-2/3 lines rise above their tier; minimum rise = 2 x LR elbow + 150 mm. Expansion loops are raised one rise above the tier (tier 2 loops drop below), nested loops step 0.9 m inwards and 0.45 m up; tier-3 loops avoid bays below the forced-draft air coolers (fan deck EL 113).
- **Expansion:** $dL = e(T) \times L$ with e from B31.3 Table C-1 (total expansion from 21 C, per class material). Rack runs are first checked for absorption by the end legs (guided cantilever, anchor at mid-run); otherwise U-loops of width 6 m (one bay) are sized: $H = \sqrt[3]{E D (dL_{loop}/2) / S_A}$, $S_A = 1.25 S_c + 0.25 S_h$.
- **Pump suctions:** shortest route, no pockets, header falling to the pumps, gate valve in the vertical leg, temporary cone strainer, eccentric reducer FLAT ON TOP at the nozzle; discharge check + block in the riser.
- **Heater transfer lines:** symmetrical multi-pass outlet manifolds (equal hydraulic length per pass), continuously rising (H-101, two-phase) or riser + continuous 1:100 fall (H-201 vacuum) to the column; 54" vacuum transfer line designed for full vacuum.
- **C-101 overhead:** self-draining at 1:200 from column to A-101 with symmetrical inlet manifold to the three bays; neutraliser/amine injection quills within 1.5 m of the column nozzle.
- **Gravity draws (C-101 to strippers):** vertical drop then continuous fall 1:100, no rising leg.
- **Steam:** branches from top of headers; lines fall to users; drip legs/traps at low points. **Relief:** PSV outlets self-draining into the flare header (falls 1:500 to D-104).

- **Spools:** shop spools <= 12 m and 3.0 x 3.0 m envelope (road transport); field welds at spool breaks, tie-ins and branch connections; high-point vents / low-point drains at every pocket.

4 Routing model summary

Level	Lines	Routed length m	Supports
header	11	1,683	261
critical	13	927	158
study	158	10,104	1734

Lines not routed at FEED (by reason):

Reason	No.	Lines (fluid-area-seq)
small bore (< 2") - field routed	13	CH-100-011, CH-100-025, BD-100-057, BFW-100-078, 1/2"-FG-100, CH-100-118, CH-100-119, SW-100-130 ...
header not routed (underground / virtual)	3	BD-100-012, BD-100-091, BD-900-018
endpoint '...' not a located nozzle/header (P&ID tie-in to another line or heater coil) -	34	P-100-016, WW-100-030, WW-100-031, P-100-046, P-100-054, MS-100-058, P-100-059, P-100-060 ...

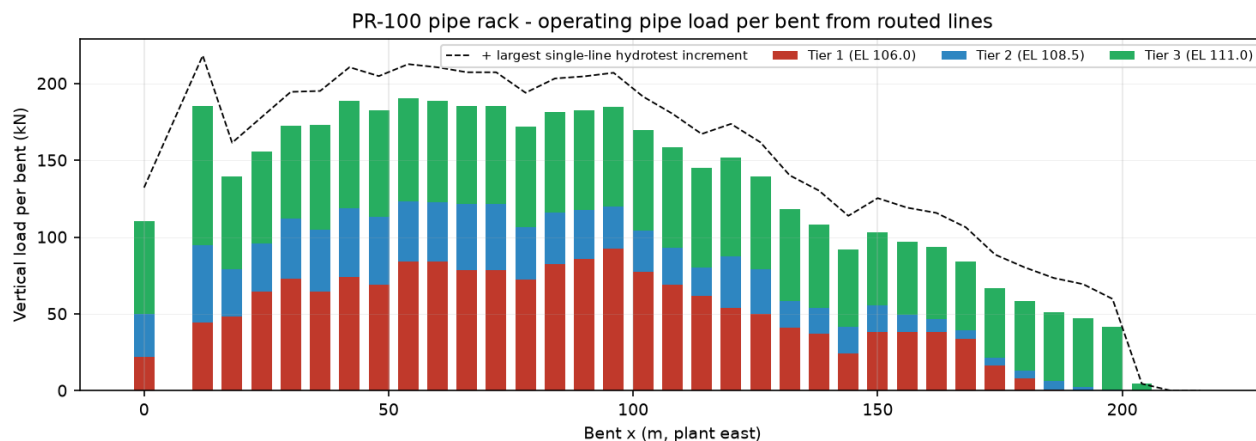
Clash check (cylinder model, bare OD): **16** line/equipment and **161** line/line interferences, of which **0 / 7** involve critical (iso) lines - all line/line items with critical lines are crossings with auto-routed study lines that are to be re-routed in the detailed 3D model; critical lines are clash-free with equipment and each other.

5 Critical lines and isometric register

Iso	Line	From -> to	Routed m	Spools	Welds S/F	Supports	Sheets
CFU-100-PI-IS O-001	28"-P-100-075-B3-H	H-101 outlet manifold -> C-101 flash zone	89.9	17	30/19	17	1
CFU-200-PI-IS O-001	54"-P-200-017-B3-H	H-201 outlet manifold -> C-201 flash zone	89.5	13	17/14	14	1
CFU-100-PI-IS O-002	36"-PG-100-10 7-A2-H	C-101 top -> A-101	85.9	11	26/13	12	1
CFU-100-PI-IS O-003	14"-P-100-084-B2-H	C-101 bottom -> P-112A/B	24.5	7	23/2	6	1
CFU-100-PI-IS O-004	10"-P-100-085-B2-H	P-112A/B -> H-201 inlet manifold	219.5	33	89/32	30	3
CFU-100-PI-IS O-005	20"-P-100-001-A1-N	TK (OSBL) -> P-101A/B	43.4	11	32/6	9	1
CFU-100-PI-IS O-006	12"-P-100-002-B1-N	P-101A/B -> E-101	65.4	12	43/10	10	1
CFU-100-PI-IS O-007	12"-P-100-050-B2-H	C-101 tray 25 -> P-108A/B	49.2	9	29/5	12	1
CFU-100-PI-IS O-008	8"-P-100-053-B1-H	E-113 -> C-101 tray 23	148.4	17	38/15	24	1
CFU-100-PI-IS O-009	10"-P-100-092-B1-H	C-101 tray 10 -> C-102	31.0	3	8/2	7	1
CFU-100-PI-IS O-010	10"-P-100-110-A2-N	D-102 -> P-103A/B	23.0	7	25/2	6	1
CFU-100-PI-IS O-011	10"-P-100-112-A2-N	D-102 -> P-104A/B	18.0	7	25/2	4	1
CFU-100-PI-IS O-012	6"-HS-100-135-S2-H	HP steam header -> E-116	39.5	5	11/6	7	1

6 Pipe-rack loading

Line weights per metre = pipe (selected schedule) + operating content (lines.json density) + insulation (thickness per temperature, 160 kg/m³ + cladding). Load per bent = w x tributary length (6 m); loops add (2H + W) w / 2 to each loop bent. Hydrotest: operating rack load + the largest single-line water-fill increment (one line tested at a time; large gas lines tested pneumatically or with temporary supports). Friction ($\mu = 0.3$) acts longitudinally at anchors/guides.



Operating pipe load per bent and tier

Tier	Max lines / bent	Max op. load kN	Equiv. UDL kN/m (10 m beam)	Max hydro increment kN	Max friction kN	Total tier kN
Tier 1 EL 106.0	16	92 @ x=96	9.2	5	20	1,706
Tier 2 EL 108.5	18	50 @ x=12	5.0	2	12	846
Tier 3 EL 111.0	13	90 @ x=12	9.0	33	27	1,957

Bent x m	Tier 1 kN	Tier 2 kN	Tier 3 kN	Total kN
12	44	50	90	185
42	74	44	70	189
54	84	39	67	191
60	84	39	66	189
66	78	43	64	185
72	78	43	64	185

Recommendation to structural: design each tier beam for the routed load above plus 25 % future space allowance and a minimum 2.0 kPa distributed load over the rack width; anchor bents to take the anchor forces from the loops (see section 7).

Rack occupancy (routed lines per tier): tier 1 20, tier 2 22, tier 3 14 (full width 10 m, 25 % spare to be held).

7 Expansion loops on the rack

Line	Tier	T C	Run m	dL mm	Loops	H m	Basis
2"-P-200-031-B 2-ST	1	410.0	177.3	834.3	2	6.5	loops raised
6"-P-200-008-B 2-ST	1	395	157.1	708.1	8	5.0	loops raised
3"-P-200-034-B 1-ST	1	395.0	136.1	690.2	0	-	end legs, mid anchor
10"-MS-900-00 9-S3-H	3	280	197.4	653.3	0	-	end legs, mid anchor
10"-P-200-005-B2-H	1	345	159.6	614.3	4	8.5	loops raised
6"-P-200-001-B 1-H	2	255	162.6	479.9	2	8.0	loops raised

Line	Tier	T C	Run m	dL mm	Loops	H m	Basis
28"-FL-900-005-F1-N	3	280.0	137.4	454.9	0	-	end legs, mid anchor
6"-P-200-041-B1-ST	2	235.0	165.0	441.6	2	7.5	loops raised
4"-P-100-105-B1-ST	1	285.0	123.0	415.9	1	8.5	loops raised
4"-P-100-049-B1-H	1	365	85.0	390.6	4	4.5	loops raised
6"-P-200-002-B1-H	2	255	131.3	387.5	2	7.0	loops raised
10"-P-100-044-B1-H	1	270	116.1	367.3	4	6.5	loops raised
6"-HS-900-008-S2-H	3	430	64.8	354.1	0	-	end legs, mid anchor
10"-P-100-085-B2-H	1	380.0	80.9	348.5	3	7.5	loops raised
6"-P-100-020-B1-H	1	325	82.5	327.4	0	-	end legs, mid anchor
4"-P-100-048-B2-H	1	365	78.8	324.0	0	-	end legs, mid anchor
6"-P-100-047-B2-H	1	325	89.8	322.1	0	-	end legs, mid anchor
18"-P-100-042-B2-H	1	305	91.0	303.2	0	-	end legs, mid anchor
10"-LS-900-010-S1-H	3	210	121.5	283.4	0	-	end legs, mid anchor
8"-P-100-045-B1-H	1	270	82.7	261.9	0	-	end legs, mid anchor
8"-P-100-051-B2-H	1	345	55.6	214.0	2	6.5	loops raised
8"-P-100-014-A1-H	2	185	102.7	205.2	1	8.5	loops raised
6"-P-100-104-A1-N	2	160.0	117.0	194.6	1	7.0	loops raised
6"-P-100-151-A1-N	2	185.0	63.0	125.8	1	6.0	loops raised

8 Stress-critical line list

Criteria (company practice, B31.3 319.4.1): **Category 1 (formal computer analysis, CAESAR II):** T > 300 C and NPS >= 6; T > 200 C and NPS >= 12; lines to rotating equipment (NPS >= 4 and T >= 150 C, or NPS >= 12); fired-heater inlet/outlet/pass lines; vacuum lines NPS >= 6; air-cooler headers (T > 120 C, NPS >= 6); NPS >= 24; flare/relief headers NPS >= 12; 600# class with T > 200 C, NPS >= 4; any routed line above 150 C failing the simplified eq.(16). **Category 2 (simplified/manual):** T > 150 C or equipment-connected NPS >= 4. **Category 3:** visual review.

Result: **Category 1: 127 lines**, Category 2: 46, Category 3: 72.

Line (Cat 1)	T C	P barg	Criteria	Model
20"-P-100-001-A1-N	60.0	3.7	rotating equipment (API 610 nozzle loads)	iso CFU-100-PI-ISO-005
12"-P-100-002-B1-N	60.0	35.0	rotating equipment (API 610 nozzle loads)	iso CFU-100-PI-ISO-006
12"-P-100-009-B1-H	165	35.0	fails simplified eq.(16)	routed
12"-P-100-013-A1-H	185	3.5	rotating equipment (API 610 nozzle loads)	routed
8"-P-100-014-A1-H	185	13.5	rotating equipment (API 610 nozzle loads)	routed
6"-P-100-016-A1-H	185	13.5	rotating equipment (API 610 nozzle loads)	-
6"-P-100-017-B1-H	240	14.5	rotating equipment (API 610 nozzle loads)	routed
6"-P-100-018-B1-H	240	14.5	air cooler header (API 661 nozzle loads)	routed
6"-P-100-019-B1-H	325	14.5	T > 300 C & NPS >= 6	routed
6"-P-100-020-B1-H	325	14.5	T > 300 C & NPS >= 6; air cooler header (API 661 nozzle loads)	routed
20"-P-100-022-B1-H	165	17.0	rotating equipment (API 610 nozzle loads)	routed
12"-P-100-023-B1-H	165.0	41.0	rotating equipment (API 610 nozzle loads)	routed
8"-P-100-024-B1-H	165.0	41.0	rotating equipment (API 610 nozzle loads)	routed
10"-FL-100-035-F1-N	230	3.5	fails simplified eq.(16)	routed
10"-FL-100-036-F1-N	230	3.5	fails simplified eq.(16)	routed
14"-P-100-038-B1-H	215	41.0	T > 200 C & NPS >= 12	routed
14"-P-100-039-B1-H	265	41.0	T > 200 C & NPS >= 12	routed
14"-P-100-040-B1-H	270	41.0	T > 200 C & NPS >= 12	routed
14"-P-100-041-C1-H	290	41.0	T > 200 C & NPS >= 12; 600# class & T > 200 C	routed
18"-P-100-042-B2-H	305	41.0	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	routed
14"-P-100-043-B1-H	270	3.5	T > 200 C & NPS >= 12; rotating equipment (API 610 nozzle loads)	routed
10"-P-100-044-B1-H	270	13.5	rotating equipment (API 610 nozzle loads)	routed
6"-P-100-047-B2-H	325	14.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	routed
4"-P-100-048-B2-H	365	14.5	rotating equipment (API 610 nozzle loads)	routed

Line (Cat 1)	T C	P barg	Criteria	Model
12"-P-100-050-B2-H	345	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; rotating equipment (API 6	iso CFU-100-PI-ISO-007
8"-P-100-051-B2-H	345	13.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	routed
12"-MS-100-058-S1-H	215	3.5	T > 200 C & NPS >= 12	-
6"-P-100-059-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-060-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-061-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-062-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-063-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-064-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-065-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-100-066-B2-H	305	38.5	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-067-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-068-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-069-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-070-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-071-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-072-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-073-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
8"-P-100-074-B3-H	545.0	18.7	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
28"-P-100-075-B3-H	545.0	18.7	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	iso CFU-100-PI-ISO-001
12"-LS-100-077-S1-H	380	6.2	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12	-

Line (Cat 1)	T C	P barg	Criteria	Model
14"-P-100-084-B2-H	390.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; rotating equipment (API 6	iso CFU-100-PI-ISO-003
10"-P-100-085-B2-H	380.0	24.0	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads); fired	iso CFU-100-PI-ISO-004
6"-P-100-086-B2-H	380.0	24.0	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	-
10"-LS-100-087-S1-H	380	6.2	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
8"-LS-100-088-S1-H	380	6.2	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
8"-P-100-089-B2-H	375	3.5	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
10"-P-100-092-B1-H	250	3.5	fails simplified eq.(16)	iso CFU-100-PI-ISO-009
10"-P-100-093-B2-H	330	3.5	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
8"-PG-100-094-B1-H	250.0	3.5	fails simplified eq.(16)	routed
8"-PG-100-095-B2-H	330.0	3.5	T > 300 C & NPS >= 6	routed
6"-PG-100-096-B2-H	375.0	3.5	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
8"-P-100-097-B1-H	250.0	3.5	rotating equipment (API 610 nozzle loads)	routed
6"-LS-100-098-S1-H	380	6.2	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
8"-P-100-099-B2-H	330.0	3.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	routed
6"-LS-100-100-S1-H	380	6.2	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
6"-P-100-101-B2-H	375.0	3.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	routed
4"-LS-100-102-S1-H	380	6.2	fails simplified eq.(16)	routed
6"-P-100-104-A1-N	160.0	14.5	air cooler header (API 661 nozzle loads)	routed
6"-FL-100-106-F1-N	330	3.5	T > 300 C & NPS >= 6; fails simplified eq.(16)	routed
36"-PG-100-107-A2-H	165.0	3.5	air cooler header (API 661 nozzle loads); NPS >= 24; fails simplified	iso CFU-100-PI-ISO-002
16"-PG-100-108-A2-N	165.0	9.7	air cooler header (API 661 nozzle loads)	routed
24"-FL-100-120-F1-N	185	3.5	NPS >= 24; flare / relief header (reaction forces)	routed
6"-PG-100-123-C1-H	235.0	13.0	air cooler header (API 661 nozzle loads); 600# class & T > 200 C	routed
12"-P-100-131-C1-H	430.0	13.0	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; 600# class & T > 200 C	routed
12"-PG-100-132-C1-H	430.0	13.0	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; 600# class & T > 200 C; f	routed
6"-P-100-133-C1-H	430.0	13.0	T > 300 C & NPS >= 6; 600# class & T > 200 C	routed
6"-P-100-134-C1-H	235.0	13.0	600# class & T > 200 C	routed

Line (Cat 1)	T C	P barg	Criteria	Model
6"-HS-100-135-S2-H	430	46.0	T > 300 C & NPS >= 6; 600# class & T > 200 C; fails simplified eq.(16)	iso CFU-100-PI-ISO-012
4"-CD-100-136-S2-P	430.0	45.0	600# class & T > 200 C	-
24"-FL-100-138-F1-N	125	3.5	NPS >= 24; flare / relief header (reaction forces)	routed
24"-FL-100-140-F1-N	290	3.5	T > 200 C & NPS >= 12; NPS >= 24; flare / relief header (reaction forc	routed
20"-PG-100-141-A1-H	185.0	3.5	air cooler header (API 661 nozzle loads)	routed
16"-P-100-147-B1-H	280.0	3.5	T > 200 C & NPS >= 12; fails simplified eq.(16)	routed
20"-P-100-148-B1-H	280.0	3.5	T > 200 C & NPS >= 12	routed
8"-P-100-149-A1-H	185.0	3.5	rotating equipment (API 610 nozzle loads)	routed
6"-P-100-150-A1-H	185.0	11.0	rotating equipment (API 610 nozzle loads); air cooler header (API 661	routed
6"-P-100-151-A1-N	185.0	11.0	air cooler header (API 661 nozzle loads)	routed
8"-MS-100-152-S3-H	280	13.0	fails simplified eq.(16)	routed
6"-CD-100-153-S1-P	215	13.0	fails simplified eq.(16)	routed
16"-FL-100-154-F1-N	130	3.5	flare / relief header (reaction forces)	routed
6"-P-200-001-B1-H	255	15.5	rotating equipment (API 610 nozzle loads)	routed
6"-P-200-002-B1-H	255	15.5	air cooler header (API 661 nozzle loads)	routed
6"-P-200-003-B2-ST	395	20.5	T > 300 C & NPS >= 6	routed
6"-P-200-004-B1-ST	310	20.5	T > 300 C & NPS >= 6	routed
10"-P-200-005-B2-H	345	15.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	routed
8"-P-200-006-B1-H	345	15.5	T > 300 C & NPS >= 6	-
6"-P-200-007-B2-H	345	15.5	T > 300 C & NPS >= 6	-
6"-P-200-008-B2-ST	395	20.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads)	routed
6"-P-200-009-B2-H	580.0	24.0	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-200-010-B2-H	580.0	24.0	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-200-011-B2-H	580.0	24.0	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
6"-P-200-012-B2-H	580.0	24.0	T > 300 C & NPS >= 6; fired heater terminal (API 560)	-
12"-P-200-013-B3-H	580.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	-
12"-P-200-014-B3-H	580.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	-

Line (Cat 1)	T C	P barg	Criteria	Model
12"-P-200-015-B3-H	580.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	-
12"-P-200-016-B3-H	580.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	-
54"-P-200-017-B3-H	580.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; fired heater terminal (AP	iso CFU-200-PI-ISO-001
3"-MS-200-018-S3-H	280	12.0	fired heater terminal (API 560)	-
48"-PG-200-024-V1-H	180	3.5	rotating equipment (API 610 nozzle loads); vacuum / external pressure;	routed
10"-P-200-025-B1-H	255.0	3.5	rotating equipment (API 610 nozzle loads)	routed
14"-P-200-027-B2-H	345.0	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; rotating equipment (API 6	routed
8"-P-200-028-B1-H	345	15.5	T > 300 C & NPS >= 6	-
4"-MS-200-032-S3-H	280	12.0	fails simplified eq.(16)	routed
10"-P-200-033-B2-ST	395.0	3.5	T > 300 C & NPS >= 6; rotating equipment (API 610 nozzle loads); fails	routed
4"-P-200-035-B2-ST	395.0	20.5	rotating equipment (API 610 nozzle loads)	routed
16"-FL-200-036-F1-N	430	3.5	T > 300 C & NPS >= 6; T > 200 C & NPS >= 12; flare / relief header (re	routed
6"-P-200-038-B1-N	255	15.5	air cooler header (API 661 nozzle loads)	-
6"-P-200-040-B1-H	345	15.5	T > 300 C & NPS >= 6; air cooler header (API 661 nozzle loads)	-
6"-P-200-041-B1-ST	235.0	15.5	air cooler header (API 661 nozzle loads)	routed
6"-MS-200-047-S3-H	280	12.0	rotating equipment (API 610 nozzle loads)	routed
2"-MS-200-049-S3-H	280	12.0	fails simplified eq.(16)	routed
42"-PG-200-050-A2-P	250.0	3.5	T > 200 C & NPS >= 12; rotating equipment (API 610 nozzle loads); vacu	routed
10"-PG-200-051-A2-P	250.0	3.5	rotating equipment (API 610 nozzle loads); vacuum / external pressure	routed
6"-PG-200-053-A2-N	100.0	3.5	vacuum / external pressure	routed
28"-FL-900-004-F1-N	280.0	3.5	T > 200 C & NPS >= 12; NPS >= 24; flare / relief header (reaction forc	routed
28"-FL-900-005-F1-N	280.0	3.5	T > 200 C & NPS >= 12; NPS >= 24; flare / relief header (reaction forc	routed
6"-HS-900-008-S2-H	430	45.5	T > 300 C & NPS >= 6; 600# class & T > 200 C; fails simplified eq.(16)	routed
10"-MS-900-009-S3-H	280	12.0	fails simplified eq.(16)	routed
10"-LS-900-010-S1-H	210	6.2	fails simplified eq.(16)	routed

Line (Cat 1)	T C	P barg	Criteria	Model
6"-MS-900-011-S3-H	280	12.0	fails simplified eq.(16)	routed
8"-CD-900-012-S1-P	215	6.7	fails simplified eq.(16)	routed
6"-BD-900-018-B1-N	395	10.0	T > 300 C & NPS >= 6	-

9 Simplified flexibility checks

ASME B31.3 eq.(16) (SI): $D y / (L - U)^2 \leq 208.3$, D = OD mm, y = resultant displacement mm (thermal growth of the straight line between terminals minus differential terminal movement - column nozzle growth from skirt base at shell design temperature), L = developed length m, U = anchor distance m. Applicable only to two-anchor systems of uniform size without intermediate restraints - lines with rack anchors/loops or failing the check go to formal analysis.

Line	Level	L m	U m	y mm	Dy/(L-U)^2	Eq.16	Cat
28"-FL-900-004-F1-N	head	71.8	58.0	193.4	725	FORMAL	Cat 1
10"-MS-900-009-S3-H	head	197.4	197.4	653.3	inf	FORMAL	Cat 1
10"-LS-900-010-S1-H	head	121.5	121.5	283.4	inf	FORMAL	Cat 1
8"-CD-900-012-S1-P	head	53.4	53.4	128.1	inf	FORMAL	Cat 1
6"-HS-900-008-S2-H	head	64.8	64.8	354.1	inf	FORMAL	Cat 1
2"-BFW-900-013-C1-H	head	47.0	47.0	71.9	inf	FORMAL	Cat 3
28"-P-100-075-B3-H	crit	61.0	46.5	309.4	1050	FORMAL	Cat 1
54"-P-200-017-B3-H	crit	71.0	54.3	380.5	1884	FORMAL	Cat 1
36"-PG-100-107-A2-H	crit	63.2	35.8	177.3	217	FORMAL	Cat 1
14"-P-100-084-B2-H	crit	19.4	13.3	56.0	524	FORMAL	Cat 1
10"-P-100-085-B2-H	crit	211.1	98.3	423.4	9	OK	Cat 1
20"-P-100-001-A1-N	crit	36.5	16.3	7.3	9	OK	Cat 1
12"-P-100-002-B1-N	crit	58.0	21.1	9.4	2	OK	Cat 1
12"-P-100-050-B2-H	crit	43.1	26.7	67.5	82	OK	Cat 1
8"-P-100-053-B1-H	crit	143.8	62.4	193.2	6	OK	Cat 2
10"-P-100-092-B1-H	crit	31.0	23.7	83.1	425	FORMAL	Cat 1
10"-P-100-110-A2-N	crit	17.3	10.4	4.8	27	OK	Cat 2
10"-P-100-112-A2-N	crit	13.8	8.0	6.2	51	OK	Cat 2
6"-HS-100-135-S2-H	crit	39.5	28.6	156.3	220	FORMAL	Cat 1
12"-P-100-008-B1-H	stud	14.2	3.7	5.7	17	OK	Cat 3
12"-P-100-009-B1-H	stud	65.5	53.7	92.9	218	FORMAL	Cat 1
8"-P-100-010-B1-H	stud	17.1	6.2	9.4	17	OK	Cat 3
12"-P-100-013-A1-H	stud	83.4	51.2	139.4	44	OK	Cat 1
8"-P-100-014-A1-H	stud	181.0	111.4	222.5	10	OK	Cat 1
8"-P-100-015-A1-H	stud	178.7	85.2	198.9	5	OK	Cat 2

Line	Level	L m	U m	y mm	Dy/(L-U)^2	Eq.16	Cat
6"-P-100-017-B 1-H	stud	156.4	89.3	245.2	9	OK	Cat 1
6"-P-100-018-B 1-H	stud	154.1	93.1	255.6	12	OK	Cat 1
6"-P-100-019-B 1-H	stud	40.4	22.5	89.4	47	OK	Cat 1
6"-P-100-020-B 1-H	stud	148.9	88.9	352.8	16	OK	Cat 1
12"-P-100-021-B 1-H	stud	22.2	11.4	17.3	48	OK	Cat 2
20"-P-100-022-B 1-H	stud	43.4	25.0	44.5	66	OK	Cat 1
12"-P-100-023-B 1-H	stud	80.3	49.5	85.6	29	OK	Cat 1
8"-P-100-024-B 1-H	stud	46.1	27.3	45.2	28	OK	Cat 1
3"-WW-100-02 8-A3-P	stud	40.4	26.6	37.4	18	OK	Cat 3
3"-WW-100-02 9-A3-P	stud	25.8	12.8	17.3	9	OK	Cat 2
3"-WW-100-03 3-A3-P	stud	31.1	16.1	19.5	8	OK	Cat 3
3"-WW-100-03 4-A3-P	stud	20.8	12.0	21.2	24	OK	Cat 3
10"-FL-100-03 5-F1-N	stud	23.2	16.5	43.0	265	FORMAL	Cat 1
10"-FL-100-03 6-F1-N	stud	33.2	26.2	68.3	385	FORMAL	Cat 1
12"-P-100-037-B 1-H	stud	17.7	7.1	15.6	46	OK	Cat 2
14"-P-100-038-B 1-H	stud	14.2	3.7	8.9	29	OK	Cat 1
14"-P-100-039-B 1-H	stud	24.7	14.1	43.5	138	OK	Cat 1
14"-P-100-040-B 1-H	stud	14.5	3.7	11.8	36	OK	Cat 1
14"-P-100-041-C 1-H	stud	49.4	29.9	103.2	96	OK	Cat 1
18"-P-100-042-B 2-H	stud	175.4	110.1	366.8	39	OK	Cat 1
14"-P-100-043-B 1-H	stud	111.8	51.2	145.9	14	OK	Cat 1
10"-P-100-044-B 1-H	stud	217.0	121.6	384.8	12	OK	Cat 1
8"-P-100-045-B 1-H	stud	164.8	90.6	267.0	11	OK	Cat 2
6"-P-100-047-B 2-H	stud	128.7	95.1	341.3	51	OK	Cat 1
4"-P-100-048-B 2-H	stud	116.8	84.1	346.0	37	OK	Cat 1
4"-P-100-049-B 1-H	stud	178.1	88.4	406.4	6	OK	Cat 2
8"-P-100-051-B 2-H	stud	113.9	61.2	235.6	19	OK	Cat 1
8"-P-100-052-B 2-H	stud	28.4	17.1	53.9	92	OK	Cat 2

Line	Level	L m	U m	y mm	Dy/(L-U)^2	Eq.16	Cat
2"-BFW-100-05 5-C1-H	stud	50.8	39.8	60.9	30	OK	Cat 3
6"-MS-100-056 -S3-H	stud	44.5	33.6	116.0	164	OK	Cat 2
10"-LS-100-08 7-S1-H	stud	30.3	25.1	139.9	1396	FORMAL	Cat 1
8"-LS-100-088- S1-H	stud	33.7	26.1	132.4	508	FORMAL	Cat 1
8"-P-100-089-B 2-H	stud	28.9	21.2	102.9	381	FORMAL	Cat 1
4"-P-100-090-B 2-H	stud	10.2	4.9	2.7	11	OK	Cat 2
10"-P-100-093- B2-H	stud	29.2	19.9	75.4	236	FORMAL	Cat 1
8"-PG-100-094 -B1-H	stud	31.5	23.2	84.3	266	FORMAL	Cat 1
8"-PG-100-095 -B2-H	stud	29.7	19.8	75.3	168	OK	Cat 1
6"-PG-100-096 -B2-H	stud	29.2	21.5	102.6	288	FORMAL	Cat 1
8"-P-100-097-B 1-H	stud	25.1	13.6	38.3	63	OK	Cat 1
6"-LS-100-098- S1-H	stud	33.7	26.1	132.4	390	FORMAL	Cat 1
8"-P-100-099-B 2-H	stud	24.7	13.6	48.9	88	OK	Cat 1
6"-LS-100-100- S1-H	stud	33.6	26.0	135.1	399	FORMAL	Cat 1
6"-P-100-101-B 2-H	stud	24.0	13.7	57.2	92	OK	Cat 1
4"-LS-100-102- S1-H	stud	33.8	26.3	137.0	274	FORMAL	Cat 1
6"-P-100-104-A 1-N	stud	142.6	117.2	195.0	51	OK	Cat 1
4"-P-100-105-B 1-ST	stud	155.8	123.4	417.1	45	OK	Cat 2
6"-FL-100-106- F1-N	stud	26.9	25.4	102.8	7721	FORMAL	Cat 1
16"-PG-100-10 8-A2-N	stud	37.0	20.7	35.9	55	OK	Cat 1
24"-FL-100-12 0-F1-N	stud	55.4	38.7	77.4	170	OK	Cat 1
6"-FL-100-121- F1-N	stud	21.0	16.8	25.7	241	FORMAL	Cat 3
6"-PG-100-123 -C1-H	stud	33.1	21.3	70.0	84	OK	Cat 1
2"-PG-100-124 -C1-H	stud	32.0	19.3	28.6	11	OK	Cat 2
12"-P-100-131- C1-H	stud	14.5	6.2	33.0	152	OK	Cat 1
12"-PG-100-13 2-C1-H	stud	11.2	6.1	32.5	413	FORMAL	Cat 1
6"-P-100-133- C1-H	stud	17.8	8.4	47.1	89	OK	Cat 1
6"-P-100-134- C1-H	stud	43.2	28.0	50.9	37	OK	Cat 1

Line	Level	L m	U m	y mm	Dy/(L-U)^2	Eq.16	Cat
24"-FL-100-140-F1-N	stud	44.2	34.5	119.1	762	FORMAL	Cat 1
20"-PG-100-141-A1-H	stud	48.4	28.4	52.2	66	OK	Cat 1
6"-PG-100-143-A1-H	stud	48.2	30.6	21.8	12	OK	Cat 2
16"-P-100-147-B1-H	stud	12.1	5.3	24.0	213	FORMAL	Cat 1
20"-P-100-148-B1-H	stud	16.3	5.3	25.1	105	OK	Cat 1
8"-P-100-149-A1-H	stud	29.8	17.5	33.9	49	OK	Cat 1
6"-P-100-150-A1-H	stud	26.2	16.5	33.0	60	OK	Cat 1
6"-P-100-151-A1-N	stud	86.9	63.4	126.6	39	OK	Cat 1
8"-MS-100-152-S3-H	stud	32.8	27.3	90.3	649	FORMAL	Cat 1
6"-CD-100-153-S1-P	stud	39.4	34.1	81.8	483	FORMAL	Cat 1
6"-P-200-001-B1-H	stud	259.7	170.2	502.4	11	OK	Cat 1
6"-P-200-002-B1-H	stud	218.3	136.0	401.2	10	OK	Cat 1
6"-P-200-003-B2-ST	stud	44.9	31.8	143.4	142	OK	Cat 1
6"-P-200-004-B1-ST	stud	20.0	8.3	31.2	39	OK	Cat 1
10"-P-200-005-B2-H	stud	265.8	165.1	635.5	17	OK	Cat 1
6"-P-200-008-B2-ST	stud	317.6	164.8	742.8	5	OK	Cat 1
48"-PG-200-024-V1-H	stud	55.4	32.7	207.9	492	FORMAL	Cat 1
10"-P-200-025-B1-H	stud	69.8	43.5	120.6	48	OK	Cat 1
14"-P-200-027-B2-H	stud	57.8	36.6	93.3	73	OK	Cat 1
2"-P-200-029-B2-H	stud	47.6	30.1	83.4	16	OK	Cat 2
3"-P-200-030-B2-H	stud	41.5	25.9	74.0	27	OK	Cat 2
2"-P-200-031-B2-ST	stud	223.7	181.7	854.7	29	OK	Cat 2
4"-MS-200-032-S3-H	stud	22.2	18.3	94.8	693	FORMAL	Cat 1
10"-P-200-033-B2-ST	stud	24.5	15.9	61.9	226	FORMAL	Cat 1
3"-P-200-034-B1-ST	stud	201.2	138.3	701.5	16	OK	Cat 2
4"-P-200-035-B2-ST	stud	33.2	17.8	65.8	32	OK	Cat 1
16"-FL-200-036-F1-N	stud	57.2	41.3	232.4	371	FORMAL	Cat 1
6"-P-200-041-B1-ST	stud	213.5	165.2	442.0	32	OK	Cat 1

Line	Level	L m	U m	y mm	Dy/(L-U)^2	Eq.16	Cat
6"-P-200-042-B 1-ST	stud	72.3	47.3	159.9	43	OK	Cat 2
2"-BFW-200-04 3-C1-H	stud	50.8	39.8	60.9	30	OK	Cat 3
6"-LS-200-044- S1-H	stud	51.7	40.5	89.0	121	OK	Cat 2
6"-MS-200-047 -S3-H	stud	25.9	18.0	59.4	160	OK	Cat 1
3"-MS-200-048 -S3-H	stud	20.9	15.8	52.2	179	OK	Cat 2
2"-MS-200-049 -S3-H	stud	17.9	15.1	50.1	400	FORMAL	Cat 1
42"-PG-200-05 0-A2-P	stud	19.2	5.6	16.0	92	OK	Cat 1
10"-PG-200-05 1-A2-P	stud	11.0	5.5	15.9	146	OK	Cat 1
3"-PG-200-052 -A2-P	stud	10.2	5.5	15.9	64	OK	Cat 2
4"-FL-200-071- F1-N	stud	44.6	34.8	53.2	63	OK	Cat 3
28"-FL-900-00 5-F1-N	stud	201.0	146.6	478.1	115	OK	Cat 1
3"-BD-900-006 -A1-N	stud	19.4	6.7	18.5	10	OK	Cat 2
6"-MS-900-011 -S3-H	stud	9.2	7.2	23.7	999	FORMAL	Cat 1

Guided-cantilever leg check used for loops / end legs: required leg L = $\sqrt{3 E D d L / S_A}$ (E cold modulus B31.3 Table C-6; S_A allowable displacement stress range).

10 Pump nozzle loads (API 610)

Pump suction/discharge piping for P-101, P-103/104, P-108 and P-112 is arranged so that the pump-side spring support carries the weight of valves/strainers, the first rigid support is adjustable, and the loop between pump and header provides flexibility. Computed nozzle loads shall not exceed API 610 Table 5 (2x for Annex F method with vendor agreement). Hot services (P-108 345 C, P-112 390 C) need warm-up bypasses and formal analysis with both pumps' operating/standby cases. Nozzle sizes below are assumed pending vendor data.

Nozzle	Fx N	Fy N	Fz N	FR N	Mx N.m	My N.m	Mz N.m	MR N.m
2"	710	580	890	1,280	460	230	350	620
3"	1,070	890	1,330	1,930	950	470	720	1,280
4"	1,420	1,160	1,780	2,560	1,330	680	1,000	1,800
6"	2,490	2,050	3,110	4,480	2,300	1,180	1,760	3,130
8"	3,780	3,110	4,890	6,920	3,530	1,760	2,580	4,710
10"	5,340	4,450	6,670	9,630	5,020	2,440	3,800	6,750
12"	6,670	5,340	8,000	11,700	6,100	2,980	4,610	8,210
14"	7,120	5,780	8,900	12,780	6,370	3,120	4,750	8,540
16"	8,450	6,670	10,230	14,850	7,320	3,660	5,420	9,890

11 Fired-heater terminals (API 560)

- H-101 / H-201 pass outlets join symmetrical manifolds on the heater north face; manifolds and the first transfer-line spool are supported from the heater structure on variable springs so that the coil terminal loads remain within API 560 Table 7 allowables (typical 8"-12" terminals: F 2.2-4.4 kN, M 1.4-3.4 kN.m; to be confirmed by heater vendor).

- Coil terminal thermal movements (heater vendor) shall be included in the transfer-line analysis; the routing gives transfer lines of 61 m (H-101) and 71 m (H-201) developed length.
- Inlet pass lines with pass flow control valves are part of the heater inlet manifold (vendor-coordinated).

12 Valve list summary

Type	Class	Isos	Study routes	P&ID est.	Total	Sizes
check	A1	0	14	2	16	2", 3", 4", 6", 8"
check	A2	0	4	0	4	2"
check	A3	0	4	0	4	3"
check	B1	2	14	0	16	6", 8", 10", 12"
check	B2	2	16	2	20	2", 4", 6", 8", 10"
check	C1	0	4	0	4	2", 3"
control	A1	0	12	2	14	1", 2", 3", 4", 6", 8", 18"
control	A2	0	4	0	4	2", 4"
control	A3	0	2	1	3	2"
control	B1	2	11	4	17	2", 3", 4", 6", 8", 10", 12", 18"
control	B2	1	5	12	18	2", 4", 6", 8", 10"
control	C1	0	6	0	6	2", 4"
control	S1	0	5	2	7	3", 4", 6", 8", 10"
control	S3	0	3	1	4	2", 3", 4", 6"
control	V1	0	1	0	1	42"
gate	A1	3	56	11	70	1", 1-1/2", 2", 3", 4", 6", 8", 10", 12"
gate	A2	6	19	0	25	2", 3", 4", 6", 10"
gate	A3	0	11	2	13	3"
gate	B1	10	48	13	71	1", 3", 4", 6", 8", 10", 12", 14", 16",
gate	B2	10	39	27	76	2", 3", 4", 6", 8", 10", 12", 14"
gate	C1	0	23	2	25	1", 2", 3", 6", 12"
gate	F1	0	13	0	13	3", 4", 6", 10", 16", 24", 28"
gate	S1	0	20	7	27	4", 6", 8", 10", 12"
gate	S2	1	1	0	2	6"
gate	S3	0	15	3	18	2", 3", 4", 6", 8", 10"
gate	U1	0	11	0	11	3", 6", 12", 18"
gate	U2	0	2	0	2	1", 1-1/2"
gate	V1	0	2	0	2	48"
gate (PSV outlet, CSO)	A1	0	0	1	1	2"
gate (PSV outlet, CSO)	S1	0	0	2	2	8", 12"
gate (vent/drain)	A1	1	50	6	57	3/4", 1"
gate (vent/drain)	A2	1	40	0	41	3/4", 1"
gate (vent/drain)	A3	0	12	4	16	3/4"
gate (vent/drain)	B1	6	72	16	94	3/4", 1"
gate (vent/drain)	B2	12	42	32	86	1", 1-1/2"
gate (vent/drain)	B3	4	0	24	28	1", 1-1/2"

Type	Class	Isos	Study routes	P&ID est.	Total	Sizes
gate (vent/drain)	C1	0	32	0	32	3/4"
gate (vent/drain)	F1	0	28	0	28	3/4"
gate (vent/drain)	S1	0	18	14	32	3/4"
gate (vent/drain)	S2	2	2	2	6	3/4", 1"
gate (vent/drain)	S3	0	16	2	18	3/4"
gate (vent/drain)	U1	0	20	0	20	3/4"
gate (vent/drain)	V1	0	2	0	2	3/4"
globe	A1	0	12	2	14	1", 2", 3", 4", 6", 8", 18"
globe	A2	0	4	0	4	2", 4"
globe	A3	0	2	1	3	2"
globe	B1	2	11	4	17	2", 3", 4", 6", 8", 10", 18"
globe	B2	1	5	12	18	2", 4", 6", 8"
globe	C1	0	6	0	6	2", 4"
globe	S1	0	5	2	7	3", 4", 6", 8", 10"
globe	S2	1	0	0	1	6"
globe	S3	0	4	1	5	2", 3", 4", 6"
globe	V1	0	1	0	1	42"
on/off (SDV)	A1	0	2	1	3	3", 8"
on/off (SDV)	C1	0	1	0	1	2"
spectacle blind	A1	1	5	1	7	1", 6", 8", 20"
spectacle blind	A2	0	1	0	1	6"
spectacle blind	A3	0	1	0	1	3"
spectacle blind	B1	0	3	3	6	1", 4", 6"
spectacle blind	C1	0	2	1	3	1", 2"
spectacle blind	F1	0	1	0	1	28"
spectacle blind	S1	0	2	0	2	8", 10"
spectacle blind	S2	0	1	0	1	6"
spectacle blind	S3	0	1	0	1	10"
spectacle blind	U1	0	2	0	2	18"
spectacle blind	U2	0	2	0	2	1", 1-1/2"
strainer (temp.)	A1	2	10	2	14	1", 2", 3", 8", 12", 20"
strainer (temp.)	A2	4	6	0	10	3", 4", 10"
strainer (temp.)	A3	0	2	0	2	3"
strainer (temp.)	B1	0	8	2	10	1", 8", 10", 14", 20"
strainer (temp.)	B2	4	10	0	14	3", 6", 8", 10", 12", 14"
strainer (temp.)	C1	0	2	0	2	6"

Control valves and orifices are supplied by I&C (tags per data/instruments.json); counts above include block/bypass valves of control stations and vent/drain valves (3/4", 1" in B2/B3).

13 Wall thickness per class (ASME B31.3 304.1.2)

$t = P D / (2 (S E W + P Y))$; $t_m = t + c$; $t_{nom} \geq t_m / 0.875$ (12.5 % mill tolerance); next standard schedule selected (ASME B36.10M), company minimum XS for $\leq 2"$ and STD for $\geq 3"$. Governing P/T = the line of that class and size with the largest required thickness. E = 1.0 seamless ($\leq 24"$) / EFW with 100 % RT; 0.85 EFW spot RT; W per Table 302.3.5 above 510 C for EFW CrMo. External pressure (vacuum) checked separately (long-cylinder buckling, FS 3).

Class A1 - 150# CS ASTM A106 Gr.B, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
2"	60.3	11.0/90.0	138.0	1.0	1.0	0.4	3.0	0.24	3.24	3.7	SCH 80	5.54
3"	88.9	12.0/80	138.0	1.0	1.0	0.4	3.0	0.39	3.39	3.87	SCH 40	5.49
4"	114.3	12.0/80	138.0	1.0	1.0	0.4	3.0	0.5	3.5	3.99	SCH 40	6.02
6"	168.3	14.5/105.0	138.0	1.0	1.0	0.4	3.0	0.88	3.88	4.43	SCH 40	7.11
8"	219.1	13.5/185	138.0	1.0	1.0	0.4	3.0	1.07	4.07	4.65	SCH 40	8.18
10"	273.0	9.7/115.0	138.0	1.0	1.0	0.4	3.0	0.96	3.96	4.52	SCH 40	9.27
12"	323.8	3.5/185	138.0	1.0	1.0	0.4	3.0	0.41	3.41	3.9	STD	9.53
20"	508.0	3.7/60.0	138.0	1.0	1.0	0.4	3.0	0.68	3.68	4.21	STD	9.53

Class A2 - 150# CS ASTM A106 Gr.B HIC-resistant (NACE MR0103), CA 6.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
2"	60.3	9.5/75	138.0	1.0	1.0	0.4	6.0	0.21	6.21	7.09	SCH 160	8.74
3"	88.9	3.5/75	138.0	1.0	1.0	0.4	6.0	0.11	6.11	6.99	SCH 80	7.62
4"	114.3	7.0/80	138.0	1.0	1.0	0.4	6.0	0.29	6.29	7.19	SCH 80	8.56
6"	168.3	3.5/75	138.0	1.0	1.0	0.4	6.0	0.21	6.21	7.1	SCH 40	7.11
10"	273.0	3.5/250.0	131.4	1.0	1.0	0.4	6.0	0.36	6.36	7.27	SCH 40 (vac: XS)	9.27
16"	406.4	9.7/165.0	138.0	1.0	1.0	0.4	6.0	1.42	7.42	8.48	STD	9.53
36"	914.0	3.5/165.0	138.0	0.85	1.0	0.4	6.0	1.36	7.36	8.41	STD	9.53
42"	1067.0	3.5/250.0	131.4	0.85	1.0	0.4	6.0	1.67	7.67	8.76	STD (vac: SCH 40)	9.53

Class A3 - 300# CS ASTM A106 Gr.B HIC-resistant (NACE MR0103), CA 6.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
3"	88.9	24.5/150	138.0	1.0	1.0	0.4	6.0	0.78	6.78	7.75	SCH 160	11.13

Class B1 - 300# CS ASTM A106 Gr.B, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
3"	88.9	20.5/395.0	93.3	1.0	1.0	0.4	3.0	0.97	3.97	4.54	SCH 40	5.49
4"	114.3	14.5/365	113.9	1.0	1.0	0.4	3.0	0.72	3.72	4.26	SCH 40	6.02
6"	168.3	20.5/310	120.2	1.0	1.0	0.4	3.0	1.43	4.43	5.06	SCH 40	7.11

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
8"	219.1	41.0/165.0	138.0	1.0	1.0	0.4	3.0	3.22	6.22	7.1	SCH 40	8.18
10"	273.0	13.5/270	128.0	1.0	1.0	0.4	3.0	1.43	4.43	5.07	SCH 40	9.27
12"	323.8	41.0/165.0	138.0	1.0	1.0	0.4	3.0	4.75	7.75	8.86	STD	9.53
14"	355.6	41.0/270	128.0	1.0	1.0	0.4	3.0	5.62	8.62	9.85	SCH 40	11.13
16"	406.4	3.5/280.0	126.1	1.0	1.0	0.4	3.0	0.56	3.56	4.07	STD	9.53
20"	508.0	17.0/165	138.0	1.0	1.0	0.4	3.0	3.11	6.11	6.99	STD	9.53

Class B2 - 300# 5Cr-1/2Mo ASTM A335 P5, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
2"	60.3	15.5/345.0	113.9	1.0	1.0	0.4	3.0	0.41	3.41	3.9	SCH 80	5.54
3"	88.9	3.5/410.0	106.6	1.0	1.0	0.4	3.0	0.15	3.15	3.6	SCH 40	5.49
4"	114.3	20.5/395.0	109.4	1.0	1.0	0.4	3.0	1.06	4.06	4.64	SCH 40	6.02
6"	168.3	24.0/580.0	24.3	1.0	1.0	0.7	3.0	7.76	10.76	12.3	SCH 120	14.27
8"	219.1	13.5/345	113.9	1.0	1.0	0.4	3.0	1.29	4.29	4.91	SCH 40	8.18
10"	273.0	24.0/380.0	111.0	1.0	1.0	0.4	3.0	2.93	5.93	6.77	SCH 40	9.27
12"	323.8	3.5/345	113.9	1.0	1.0	0.4	3.0	0.5	3.5	4.0	STD	9.53
14"	355.6	3.5/390.0	110.0	1.0	1.0	0.4	3.0	0.57	3.57	4.07	STD	9.53
18"	457.0	41.0/305	116.4	1.0	1.0	0.4	3.0	7.94	10.94	12.5	XS	12.7

Class B3 - 300# 9Cr-1Mo ASTM A335 P9, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
8"	219.1	18.7/545.0	47.0	1.0	1.0	0.7	3.0	4.24	7.24	8.27	SCH 60	10.31
12"	323.8	3.5/580.0	28.8	1.0	1.0	0.7	3.0	1.95	4.95	5.66	STD (vac: STD)	9.53
28"	711.0	18.7/545.0	47.0	1.0	0.923	0.7	3.0	14.88	17.88	20.43	WT 20.62	20.62
54"	1372.0	3.5/580.0	28.8	1.0	0.846	0.7	3.0	9.76	12.76	14.59	WT 15.88 (vac: WT 19.05)	15.88

Class C1 - 600# CS killed ASTM A106 Gr.B, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
2"	60.3	55.0/150	138.0	1.0	1.0	0.4	3.0	1.18	4.18	4.78	SCH 80	5.54
3"	88.9	25.0/75	138.0	1.0	1.0	0.4	3.0	0.8	3.8	4.34	SCH 40	5.49
6"	168.3	13.0/430.0	72.4	1.0	1.0	0.4	3.0	1.5	4.5	5.14	SCH 40	7.11
12"	323.8	13.0/430.0	72.4	1.0	1.0	0.4	3.0	2.88	5.88	6.73	STD	9.53

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
14"	355.6	41.0/290	124.1	1.0	1.0	0.4	3.0	5.8	8.8	10.05	SCH 40	11.13

Class S1 - 150# CS ASTM A106 Gr.B, CA 1.5 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
4"	114.3	6.2/380	105.6	1.0	1.0	0.4	1.5	0.33	1.83	2.1	SCH 40	6.02
6"	168.3	13.0/215	136.4	1.0	1.0	0.4	1.5	0.8	2.3	2.63	SCH 40	7.11
8"	219.1	6.2/380	105.6	1.0	1.0	0.4	1.5	0.64	2.14	2.45	SCH 40	8.18
10"	273.0	6.2/380	105.6	1.0	1.0	0.4	1.5	0.8	2.3	2.63	SCH 40	9.27
12"	323.8	6.2/380	105.6	1.0	1.0	0.4	1.5	0.95	2.45	2.8	STD	9.53

Class S2 - 600# 1.25Cr-1/2Mo ASTM A335 P11, CA 1.5 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
4"	114.3	45.0/430.0	117.4	1.0	1.0	0.4	1.5	2.16	3.66	4.18	SCH 40	6.02
6"	168.3	46.0/430	117.4	1.0	1.0	0.4	1.5	3.25	4.75	5.42	SCH 40	7.11

Class S3 - 300# CS ASTM A106 Gr.B, CA 1.5 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
2"	60.3	12.0/280	126.1	1.0	1.0	0.4	1.5	0.29	1.79	2.04	SCH 80	5.54
3"	88.9	12.0/280	126.1	1.0	1.0	0.4	1.5	0.42	1.92	2.2	SCH 40	5.49
4"	114.3	12.0/280	126.1	1.0	1.0	0.4	1.5	0.54	2.04	2.33	SCH 40	6.02
6"	168.3	13.0/290.0	124.1	1.0	1.0	0.4	1.5	0.88	2.38	2.72	SCH 40	7.11
8"	219.1	13.0/280	126.1	1.0	1.0	0.4	1.5	1.12	2.62	3.0	SCH 40	8.18
10"	273.0	12.0/280	126.1	1.0	1.0	0.4	1.5	1.29	2.79	3.19	SCH 40	9.27

Class U1 - 150# CS ASTM A106 Gr.B (cement-lined CW >= 12in), CA 1.5 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
3"	88.9	8.2/60	138.0	1.0	1.0	0.4	1.5	0.26	1.76	2.02	SCH 40	5.49
6"	168.3	8.2/60	138.0	1.0	1.0	0.4	1.5	0.5	2.0	2.28	SCH 40	7.11
12"	323.8	8.2/60	138.0	1.0	1.0	0.4	1.5	0.96	2.46	2.81	STD	9.53
18"	457.0	8.2/60	138.0	1.0	1.0	0.4	1.5	1.35	2.85	3.26	STD	9.53

Class V1 - 150# 5Cr-1/2Mo ASTM A335 P5 / A691 5CR, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
48"	1219.0	3.5/180	119.4	1.0	1.0	0.4	3.0	1.78	4.78	5.47	STD (vac: WT 15.88)	9.53

Class F1 - 150# CS killed, impact-tested ASTM A333 Gr.6, CA 3.0 mm

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selecte d	WT mm
3"	88.9	3.5/90	138.0	1.0	1.0	0.4	3.0	0.11	3.11	3.56	SCH 40	5.49
4"	114.3	3.5/150	138.0	1.0	1.0	0.4	3.0	0.14	3.14	3.59	SCH 40	6.02
6"	168.3	3.5/330	118.0	1.0	1.0	0.4	3.0	0.25	3.25	3.71	SCH 40	7.11

NPS	OD mm	P/T gov.	S MPa	E	W	Y	c	t mm	tm mm	t_nom req	Selected	WT mm
10"	273.0	3.5/230	134.3	1.0	1.0	0.4	3.0	0.36	3.36	3.83	SCH 40	9.27
16"	406.4	3.5/430	72.4	1.0	1.0	0.4	3.0	0.98	3.98	4.55	STD	9.53
24"	610.0	3.5/290	124.1	1.0	1.0	0.4	3.0	0.86	3.86	4.41	STD	9.53
28"	711.0	3.5/280.0	126.1	0.85	1.0	0.4	3.0	1.16	4.16	4.75	STD	9.53

14 Upstream data issues and assumptions

Issues for the owning disciplines (not changed by piping):

- 10"-P-100-044-B1-H: loop leg 6.5 m exceeds rack width - rack extension or bellows required
- 10"-P-200-005-B2-H: loop leg 8.5 m exceeds rack width - rack extension or bellows required
- 25 shell & tube exchangers have bottom nozzles below EL 101.2 (E-115A, E-115B, E-116, E-114, E-106A, E-106B, E-107, E-108A...): insufficient room for bottom elbows/valves/drains - recommend exchanger supports raised to $\geq$ EL 101.5 (layout)
- H-201 -> C-201 vacuum transfer line developed length 71 m (heater south of rack, column north): longer than typical (< 40 m); process to confirm flash-zone pressure drop / consider relocating H-201 adjacent to C-201 (layout)
- H-101 -> C-101 transfer line 61 m with rack crossing above tier 3 - dedicated support steel at bents x=96/102 required (structural)
- Pump nozzle sizes not in mech.json - assumed 1-2 sizes below line size for suction reducers (vendor data)
- C-101 kero/diesel/AGO draw nozzles all at the same azimuth (layout.json): gravity drops must be staggered - recommend re-orienting draw nozzles in 20-30 deg steps (mechanical / layout)

Assumptions made by the routing model:

- PSV-1010 outlet assumed at P-101A (15.0, 81.5, EL 102.80) - relief line routed self-draining to flare header
- PSV-1002 outlet assumed at D-101A (37.0, 54.9, EL 106.30) - relief line routed self-draining to flare header
- PSV-1003 outlet assumed at D-101B (37.0, 44.9, EL 106.30) - relief line routed self-draining to flare header
- PSV-1009 outlet assumed at C-102 (108.4, 96.0, EL 112.90) - relief line routed self-draining to flare header
- PSV-1001 outlet assumed at C-101 (98.2, 93.3, EL 142.70) - relief line routed self-draining to flare header
- PSV-1004 outlet assumed at D-102 (80.0, 87.3, EL 109.40) - relief line routed self-draining to flare header
- PSV-1005 outlet assumed at C-105 (66.0, 97.8, EL 121.70) - relief line routed self-draining to flare header
- PSV-1006 outlet assumed at D-105 (66.5, 88.7, EL 106.70) - relief line routed self-draining to flare header
- PSV-1008 outlet assumed at E-116 (65.0, 104.0, EL 102.95) - relief line routed self-draining to flare header
- PSV-1007 outlet assumed at C-106 (54.3, 97.2, EL 135.20) - relief line routed self-draining to flare header
- PSV-2001 outlet assumed at C-201 (178.2, 93.5, EL 145.70) - relief line routed self-draining to flare header
- PSV-2002 outlet assumed at D-201 (197.0, 104.3, EL 102.90) - relief line routed self-draining to flare header
- H-101 pass-outlet stubs assumed on north face at 2.2 m pitch, EL 103.000 (heater vendor to confirm terminal points)
- H-201 pass-outlet stubs assumed on north face at 3.0 m pitch, EL 103.000 (heater vendor to confirm terminal points)
- D-102 second hydrocarbon outlet nozzle assumed 3.0 m west of 'liquid_outlet' (layout.json has one; P&ID has separate D-102 to P-103 and D-102 to P-104 lines)
- P-101A nozzle 'discharge' shared by 12"-P-100-002-B1-N and 8"-P-100-003-B1-N - separate nozzle assumed
- P-101B nozzle 'discharge' shared by 12"-P-100-002-B1-N and 8"-P-100-003-B1-N - separate nozzle assumed
- PSV-1011 outlet assumed at E-113 (41.5, 107.5, EL 102.95) - relief line routed self-draining to flare header
- C-101 nozzle 'feed' shared by 28"-P-100-075-B3-H and 4"-P-100-090-B2-H - separate nozzle assumed
- C-101 nozzle 'bottoms' shared by 14"-P-100-084-B2-H and 4"-BD-100-091-B2-N - separate nozzle assumed
- H-201 nozzle 'inlet' shared by 10"-P-100-085-B2-H and 6"-P-200-009-B2-H - separate nozzle assumed
- PSV-2003 outlet assumed at E-201 (45.0, 107.5, EL 102.95) - relief line routed self-draining to flare header

15 Deliverables

- data/routing.json - routing model (polylines, fittings, supports, welds, spools, lengths, rack runs, loops, checks)
- 3d/CFU-000-PI-3DM-002_Piping.glb + 3d/viewer-piping.html (equipment + piping, toggle, click-to-identify)
- piping/iso/CFU-xxx-PI-IS0-0nn_*.pdf/.svg and merged set CFU-000-PI-IS0-000_Isometric-Set.pdf
- piping/CFU-000-PI-MT0-001_Piping-MT0.xlsx (pipe, fittings, flanges, valves, welds/spools, supports, line cross-check)